



# NORTH SYDNEY BOYS HIGH SCHOOL

## 2026 HSC ASSESSMENT TASK 3

# Mathematics Extension 1

### General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using blue or black pen
- Calculators approved by NESA may be used
- A separate reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Student Number and select your Class teacher on the front of each booklet

(Please tick or highlight)

- ☐ Ms Sarofim
- ☐ Ms Everingham
- ☐ Ms Cai
- ☐ Mr Berry
- ☐ Dr Vranešević
- ☐ Ms Moss
- ☐ Ms Lee

**Class Teacher:**

Student Number:

(To be used by the exam markers only.)

| Question No | 1-10            | 11              | 12              | 13              | 14              | Total           | Total            |
|-------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|
| Mark        | $\overline{10}$ | $\overline{15}$ | $\overline{15}$ | $\overline{15}$ | $\overline{15}$ | $\overline{70}$ | $\overline{100}$ |

## Section I: Multiple Choice

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

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1 Find the value of  $p$  so that the vectors  $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$  and  $\begin{pmatrix} p \\ 20 \end{pmatrix}$  are parallel.

A.  $-8$

B.  $-4$

C.  $4$

D.  $50$

2 The roots of the polynomial  $x^3 + 4x^2 - 3x + 2 = 0$  are  $\alpha, \beta$  and  $\gamma$ . Evaluate

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

A.  $-\frac{2}{3}$

B.  $-\frac{3}{4}$

C.  $\frac{2}{3}$

D.  $\frac{3}{2}$

3 Solve

$$\frac{3}{x+2} > 1$$

A.  $(1, \infty)$

B.  $(-2, 1)$

C.  $(-\infty, -2) \cup (-2, \infty)$

D.  $(-\infty, -2) \cup (1, \infty)$

4 Which of the following is an expression for  $a \cos x - b \sin x$ ?

A.  $\sqrt{a^2 + b^2} \cos \left( x + \tan^{-1} \frac{a}{b} \right)$

B.  $\sqrt{a^2 + b^2} \cos \left( x - \tan^{-1} \frac{a}{b} \right)$

C.  $\sqrt{a^2 + b^2} \cos \left( x + \tan^{-1} \frac{b}{a} \right)$

D.  $\sqrt{a^2 + b^2} \cos \left( x - \tan^{-1} \frac{b}{a} \right)$

5 Find

$$\int \frac{\cos x}{1 + \sin^2 x} dx$$

A.  $\tan^{-1}(\sin x) + C$

B.  $-\frac{1}{1 + \sin x} + C$

C.  $\ln|1 + \sin^2 x| + C$

D.  $2 \ln |\sin x| + C$

6  $P(x)$  is a polynomial with a factor of  $(x + 1)$  and a remainder of 8 when it is divided by  $(2x + 1)$ . What is the remainder when the quotient of  $P(x)$  and  $(x + 1)$  is divided by  $(2x + 1)$ ?

A.  $-16$

B.  $-4$

C.  $4$

D.  $16$

- 7  $f(x)$  is a one-to-one function which passes through the point  $(a, b)$ .  
Let  $g(x) = f^{-1}(x)$ .  
What is the gradient of the tangent to  $g(x)$  at the point  $(b, a)$ ?

A.  $\frac{1}{f'[g(a)]}$

B.  $\frac{1}{f'[g(b)]}$

C.  $\frac{1}{g'[f(a)]}$

D.  $\frac{1}{g'[f(b)]}$

- 8 What is the constant term in the expansion of  $(2 + x) \left(4x + \frac{1}{x^2}\right)^8$ ?

A.  $2^{10} \binom{8}{3}$

B.  $2^3 \binom{8}{1} + \binom{8}{2}$

C.  $2^4 \binom{8}{4}$

D. There is no constant term

- 9 Evaluate the integral

$$\int_{-\frac{3\pi}{2}}^{\frac{3\pi}{2}} \cos^{-1}(\sin x) dx$$

A. 0

B.  $3\pi$

C.  $\frac{3\pi^2}{2}$

D.  $\frac{9\pi^2}{4}$

**10**

How many ways can the letters from FLUFFBALL be arranged such that there are no identical letters in adjacent positions?

A. 2040

B. 5670

C. 6480

D. 9960

## Section II

**60 marks**

**Attempt Questions 11-14**

**Allow about 1 hour 45 minutes for this section**

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

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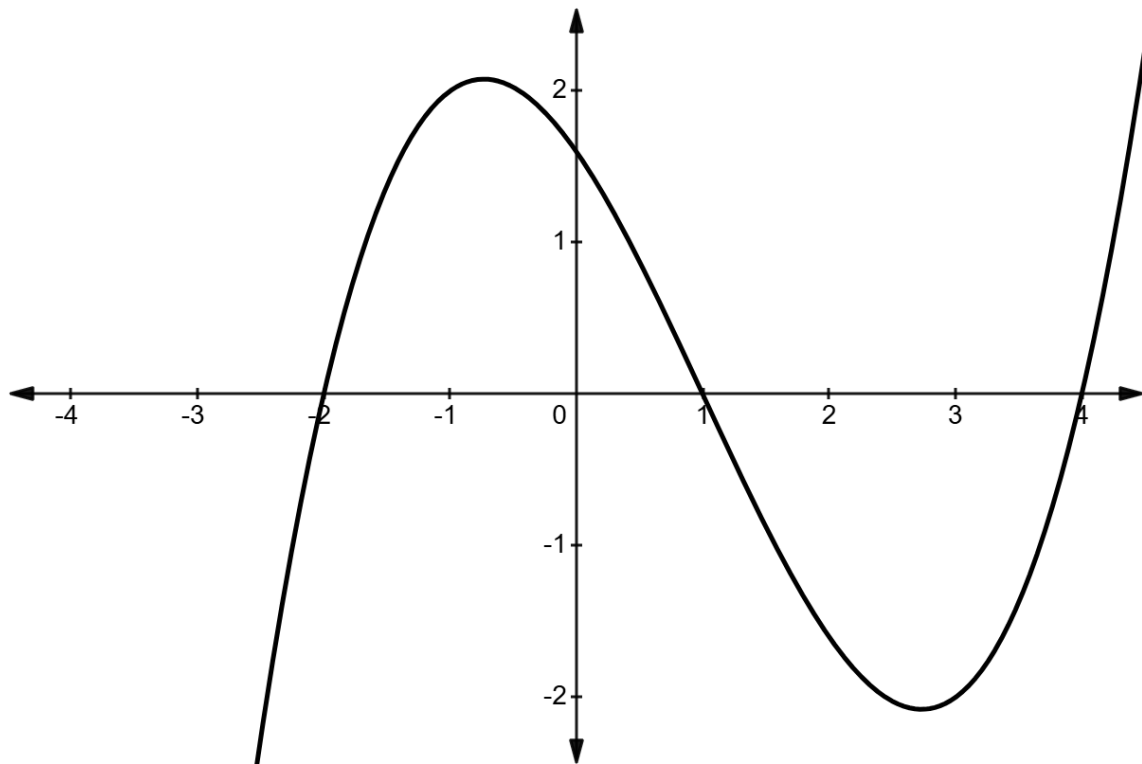
**Question 11** (15 Marks) Use the Question 11 Writing Booklet

- (a) Solve  $|2x + 5| > 5$  **2**
- (b) Two vectors are defined as  $\vec{u} = 2\vec{i} + 3\vec{j}$  and  $\vec{v} = \vec{i} - 2\vec{j}$ . Find
- (i)  $\vec{u} + \vec{v}$  **1**
- (ii)  $\vec{u} \cdot \vec{v}$  **1**
- (iii)  $|2\vec{u} - 3\vec{v}|$  **2**
- (c) Use the substitution  $t = \tan \theta$  to solve  $\cos 2\theta + \tan \theta = 1$  for  $0 \leq \theta \leq 2\pi$  **3**
- (d) The polynomial  $f(x) = x^3 + ax^2 + bx + 9$  has a double root at  $x = -3$ . Find the values of  $a$  and  $b$ . **3**
- (e) Use the substitution  $u^2 = x + 1$  to evaluate the integral **3**

$$\int_3^8 \frac{x}{\sqrt{x+1}} dx$$

**Question 12** (15 Marks) Use the Question 12 Writing Booklet

- (a) The graph  $y = f(x)$  is shown below

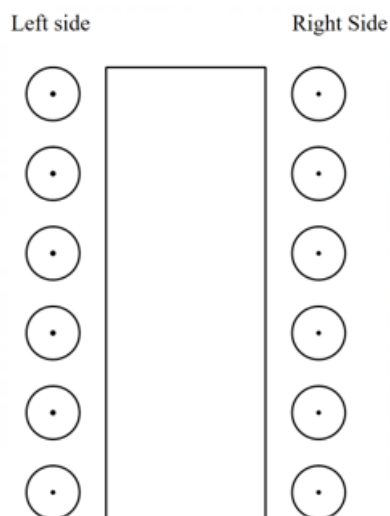


Use the axes provided in your Question 12 Writing Booklet to sketch the following

- |       |                      |   |
|-------|----------------------|---|
| (i)   | $y =  f(x) $         | 1 |
| (ii)  | $y = f( x )$         | 1 |
| (iii) | $y = \frac{1}{f(x)}$ | 2 |
| (iv)  | $y^2 = f(x)$         | 3 |

**Question 12 continues on page 8**

- (b) At a tribunal, 12 guests are to be seated on 2 sides of a rectangular table with 6 chairs on each side as shown in the diagram below. **2**



Harry and Kim need to sit on the opposite side to Evrart, Joyce and Titus. In how many ways can they be seated?

- (c) (i) Sketch the graph of  $y = 2 \cos^{-1} x$  clearly showing all intercepts and endpoints **1**
- (ii) Find the area bounded by the curve  $y = 2\cos^{-1} x$ , the  $x$ -axis and the  $y$ -axis in the first quadrant **2**
- (iii) The area in part (ii) is rotated about the  $y$ -axis. Find the exact volume of the solid formed **3**

**End of Question 12**

**Question 13** (15 Marks) Use the Question 13 Writing Booklet

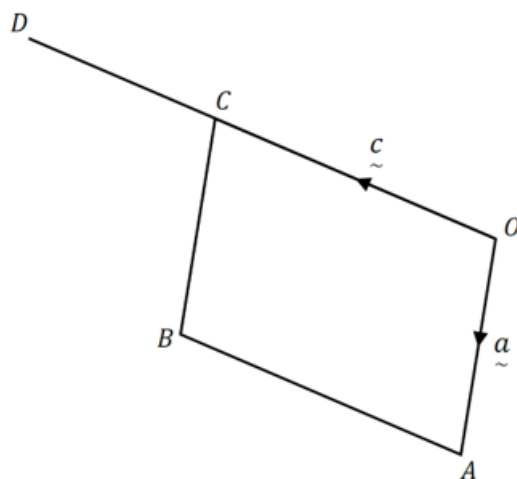
- (a) Prove by mathematical induction that  $n^3 + (n + 1)^3 + (n + 2)^3$  is divisible by 9 for integers  $n \geq 1$ . 3

- (b) (i) Solve  $x^3 - 4x^2 + 5x - 2 = 0$  for  $x$  2

- (ii) Express  $2 \sin^2 \theta - \sin \theta \sec \theta = 2 \sin 2\theta - 2$  in terms of  $\tan \theta$  2

- (iii) Hence or otherwise solve  $2 \sin^2 \theta - \sin \theta \sec \theta = 2 \sin 2\theta - 2$  for  $0 \leq \theta \leq \pi$  1

- (c) In the parallelogram  $OABC$ ,  $\overrightarrow{OA} = \underline{\underline{a}}$  and  $\overrightarrow{OC} = \underline{\underline{c}}$ .



$X$  is the midpoint of the interval  $AC$ , and  $OCD$  is a straight line such that the ratio  $OC:CD = k:1$

- (i) Find  $\overrightarrow{OX}$  in terms of  $\underline{\underline{a}}$  and  $\underline{\underline{c}}$  1

- (ii) Write an expression for  $\overrightarrow{OD}$  in terms of  $\underline{\underline{c}}$  1

- (iii) Given  $\overrightarrow{XD} = 3\underline{\underline{c}} - \frac{1}{2}\underline{\underline{a}}$ , find the value of  $k$  2

- (d) (i) Show that 1

$$\binom{n+1}{k+1} = \frac{n+1}{k+1} \binom{n}{k}$$

- (ii) Hence or otherwise show that 2

$$\binom{n}{0} + \frac{1}{2}\binom{n}{1}x + \frac{1}{3}\binom{n}{2}x^2 + \frac{1}{4}\binom{n}{3}x^3 + \dots + \frac{1}{n+1}\binom{n}{n}x^n = \frac{(1+x)^{n+1} - 1}{x(n+1)}$$

**Question 14** (15 Marks) Use the Question 14 Writing Booklet

- (a) A set contains  $n$  positive integers greater than 1. The prime factors of each integer in the set are less than 10. Find the minimum value of  $n$  such that the set must contain a pair of integers whose product is a perfect square. 2

- (b) A particle is launched from the origin with a speed of  $U$  at an angle of  $\theta$  to the horizontal. It passes through point  $P(p, q)$  with a speed of  $W$ .

The velocity vector of the particle as it passes through  $P$  is perpendicular to the velocity vector of when the particle was launched from the origin.

There is a constant acceleration due to gravity of  $g$ .

The equations of motion are given by

$$\underline{\underline{a}} = -g\underline{\underline{j}}$$

$$\underline{\underline{v}} = U \cos \theta \underline{\underline{i}} + (U \sin \theta - gt) \underline{\underline{j}}$$

$$\underline{\underline{r}} = Ut \cos \theta \underline{\underline{i}} + \left( Ut \sin \theta - \frac{1}{2}gt^2 \right) \underline{\underline{j}}$$

(do NOT prove these).

- (i) Show that the time when the particle passes through point  $P$  is given by 2

$$t = \frac{U}{g \sin \theta}$$

- (ii) Show that the launch speed of the particle is  $U = \sqrt{pg \tan \theta}$  1

- (iii) Prove that the product of  $U$  and  $W$  is independent of  $q$  and  $\theta$  3

- (c) The hands of an analogue clock are  $OA$  and  $OB$ , where  $OA = 8$  cm and  $OB = 15$  cm,  $O$  is the origin and  $t \geq 0$  is the number of minutes past midnight. 3

There is an angle of  $\theta$  between the hands at the origin

How fast, in cm/min, is the distance between the tips of the hands changing at 3:00?  
Give your answer as an exact value.

**Question 14 continues on page 11**

- (d) The polynomial, with real coefficients,  $P(x) = x^4 + ax^3 + bx^2 + ax + c$  has exactly three distinct real roots at  $\tan \theta$ ,  $\tan 2\theta$  and  $\tan 3\theta$ . 4

Find the smallest positive value for  $\theta$ .

**End of paper**

$$Q1./ \quad \frac{2}{-5} = \frac{p}{20}$$

$$-5p = 40$$

$$p = -8$$

$\therefore$  (A)

$$Q2./ \quad \frac{1}{a} + \frac{1}{b} + \frac{1}{r} = \frac{Br + ar + aB}{abr}$$

$$= \frac{-3/1}{-2/1}$$

$$= \frac{3}{2}$$

$\therefore$  (D)

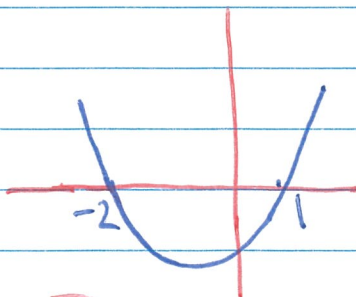
$$Q3./ \quad \frac{3}{(x+2)} > 1$$

$$3(x+2) > (x+2)^2$$

$$(x+2)^2 - 3(x+2) < 0$$

$$(x+2)[(x+2) - 3] < 0$$

$$(x+2)(x-1) < 0$$



$\therefore$  (B)

Q4./  $r \cos(x + \theta) = a \cos x - b \sin x$

$$r \cos x \cos \theta - r \sin x \sin \theta = a \cos x - b \sin x$$

$$\therefore r \cos \theta = a \text{ (1)}, r \sin \theta = b \text{ (2)}$$

$$\text{(2)} \div \text{(1)}$$

$$\frac{r \sin \theta}{r \cos \theta} = \frac{b}{a}$$

$$\tan \theta = \frac{b}{a}$$

$$\theta = \tan^{-1}\left(\frac{b}{a}\right)$$

$$\text{(1)}^2 + \text{(2)}^2$$

$$(r \cos \theta)^2 + (r \sin \theta)^2 = a^2 + b^2$$

$$r^2 (\cos^2 \theta + \sin^2 \theta) = a^2 + b^2$$

$$\therefore r = \sqrt{a^2 + b^2}$$

$$\therefore a \cos x - b \sin x = \sqrt{a^2 + b^2} \cos\left(x + \tan^{-1}\left(\frac{b}{a}\right)\right)$$

$\therefore$  (C)

Q5./  $\int \frac{\overset{f'(x)}{\cos x}}{1 + \sin x} dx = \tan^{-1}(\sin x) + C$

$$\uparrow$$

$$[f(x)]^2$$

$\therefore$  (A)

Q6./  $P(x) = (x+1) Q(x)$

$$P(-1/2) = 8$$

$$\therefore 8 = (-1/2 + 1) Q(-1/2)$$

$$Q(-1/2) = \frac{8}{(1/2)}$$

$$= 16$$

$\therefore$  (D)

Q7. /  $f[g(x)] = x$

$$g'(x) f'(g(x)) = 1$$

$$\therefore g'(x) = \frac{1}{f'[g(x)]}$$

$$g'(b) = \frac{1}{f'[g(b)]}$$

$\therefore$  (B)

Q8. /  $t_{k,r} = \binom{3}{k} 2^{3-k} x^k \binom{6}{r} x^{6-r} (x^{-2})^r$   
 $= \binom{3}{k} \binom{6}{r} 2^{3-k} x^{6+k-3r}$

$$\therefore k - 3r = -6$$

$$3r - k = 6$$

$$3r = 6 + k$$

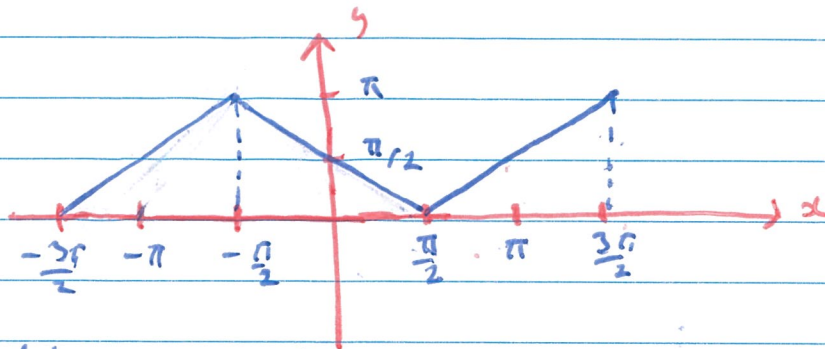
| k | r   |   |
|---|-----|---|
| 0 | 2   | ✓ |
| 1 | 7/3 | ✗ |
| 2 | 8/3 | ✗ |
| 3 | 3   | ✓ |

$$\therefore \text{constant} = \binom{3}{0} \binom{6}{2} 2^{3-0} + \binom{3}{3} \binom{6}{3} 2^{3-3}$$

$$= 8 \binom{6}{2} + \binom{6}{3}$$

$\therefore$  (A)

Q9./  $\int_{-\frac{3\pi}{2}}^{\frac{3\pi}{2}} \cos^{-1} \sin x \, dx = \int_{-\frac{3\pi}{2}}^{\frac{3\pi}{2}} \cos^{-1} [\cos(\frac{\pi}{2} - x)] \, dx$



$3 \times \left( \frac{1}{2} \times \pi \times \pi \right)$   
↑ base ↑ height

$= \frac{3\pi^2}{2}$

$\therefore$  (C)

Q10./

$3! \left[ \binom{4}{1} \binom{5}{1} + \binom{4}{1} \binom{3}{1} \binom{6}{2} + \binom{4}{3} \binom{7}{3} \right]$

↑ arrange UBA     
 ↑ FLFLF     
 ↑ L     
 ↑ FLF     
 ↑ F     
 ↑ LL     
 ↑ FFF     
 ↑ LLL

$= 2040$

$\therefore$  (A)

## Detailed explanation to Q10

Begin by arranging the non repeated letters.  
3! ways

   U    B    A   

notice there are 4 'gaps'

Case 1 (FFF all in the same gap)

   F L F L F    U    B    A   

4  
↑  
choose gap

x

5  
↑  
choose position  
for extra L

(2 L's must go  
between F's)

Case 2 (FF together, F apart)

   F L F    U    F    B    A   

4  
↑  
choose FF  
gap

x

3  
↑  
choose F  
gap

x

(6)  
(2)  
↑  
choose  
position for  
two extra L's

(one L must  
go between F's)

Case 3 (All F's apart)

   F    U    F    B    F    A   

4 C<sub>3</sub>  
↑  
choose gaps  
for F's

x

7 C<sub>2</sub>  
↑  
choose gaps  
for L's

## Question 11

(a)  $|2x + 5| > 5$

$$\begin{aligned} \therefore 2x + 5 > 5 & \text{ or } 2x + 5 < -5 \\ 2x > 0 & \text{ or } 2x < -10 \\ x > 0 & \text{ or } x < -5 \end{aligned}$$

(b)(i)  $2\hat{i} + 3\hat{j} + \hat{i} - 2\hat{j} = 3\hat{i} + \hat{j}$

(ii)  $(2)(1) + (3)(-2) = 2 - 6$   
 $= -4$

(iii)  $4\hat{i} + 6\hat{j} - (3\hat{i} - 6\hat{j}) = \hat{i} + 12\hat{j}$

$$\begin{aligned} |2\hat{i} - 3\hat{j}| &= \sqrt{1^2 + 12^2} \\ &= \sqrt{145} \end{aligned}$$

(c)  $\cos 2\theta + \tan \theta = 1$

$$t = \tan \theta$$

$$\frac{1-t^2}{1+t^2} + t = 1$$

$$\therefore 1-t^2 + t(1+t^2) = 1+t^2$$

$$1-t^2 + t + t^3 = 1+t^2$$

$$t^3 - 2t + t = 0$$

$$t(t^2 - 2t + 1) = 0$$

$$t(t-1)^2 = 0$$

$$\therefore t = 0 \text{ or } t = 1$$

$$\tan \theta = 0 \text{ or } \tan \theta = 1$$

$$\therefore \theta = 0, \pi, 2\pi, \pi/4 \text{ or } 5\pi/4$$

$$\begin{aligned}
 \text{(d)} \quad f(-3) &= (-3)^3 + a(-3)^2 + b(-3) + 9 \\
 &= -27 + 9a - 3b + 9 \\
 &= 9a - 3b - 18 \\
 &= 0
 \end{aligned}$$

$$\therefore 3a - b - 6 = 0$$

$$b = 3a - 6 \dots (1)$$

$$f'(x) = 3x^2 + 2ax + b$$

$$f'(-3) = 0$$

$$\therefore 3(-3)^2 + 2a(-3) + b = 0$$

$$27 - 6a + b = 0 \dots (2)$$

$$(1) \rightarrow (2)$$

$$27 - 6a + 3a - 6 = 0$$

$$21 - 3a = 0$$

$$3a = 21$$

$$a = 7$$

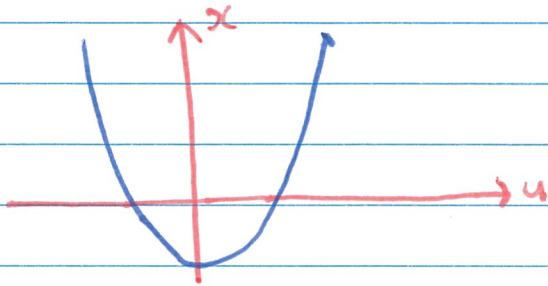
$$\text{sub} \rightarrow (1)$$

$$b = 3(7) - 6$$

$$b = 15$$

$$(e) \quad u^2 = x + 1$$

$$x = u^2 - 1$$



restrict to  $u \geq 0$  so function is one-one

$$\frac{dx}{du} = 2u$$

$$dx = 2u \, du$$

$$\text{when } x = 3 \quad u^2 = 4$$

$$u = 2$$

$$\text{when } x = 8 \quad u^2 = 9$$

$$u = 3$$

$$\therefore \int_3^8 \frac{x}{\sqrt{x+1}} \, dx = \int_2^3 \frac{u^2 - 1}{u} 2u \, du$$

$$= \int_2^3 (2u^2 - 2) \, du$$

$$= \left[ \frac{2u^3}{3} - 2u \right]_2^3$$

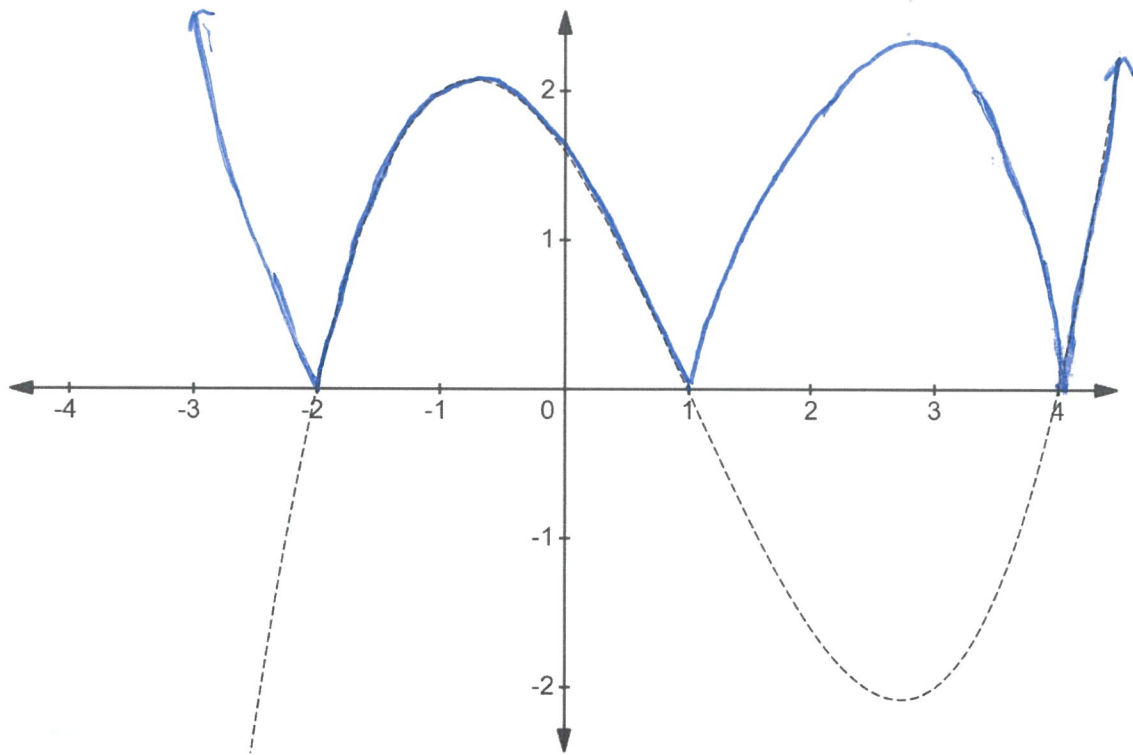
$$= \left( \frac{2 \times 3^3}{3} - 2 \times 3 \right) - \left( \frac{2 \times 2^3}{3} - 2^2 \right)$$

$$= 12 - \frac{4}{3}$$

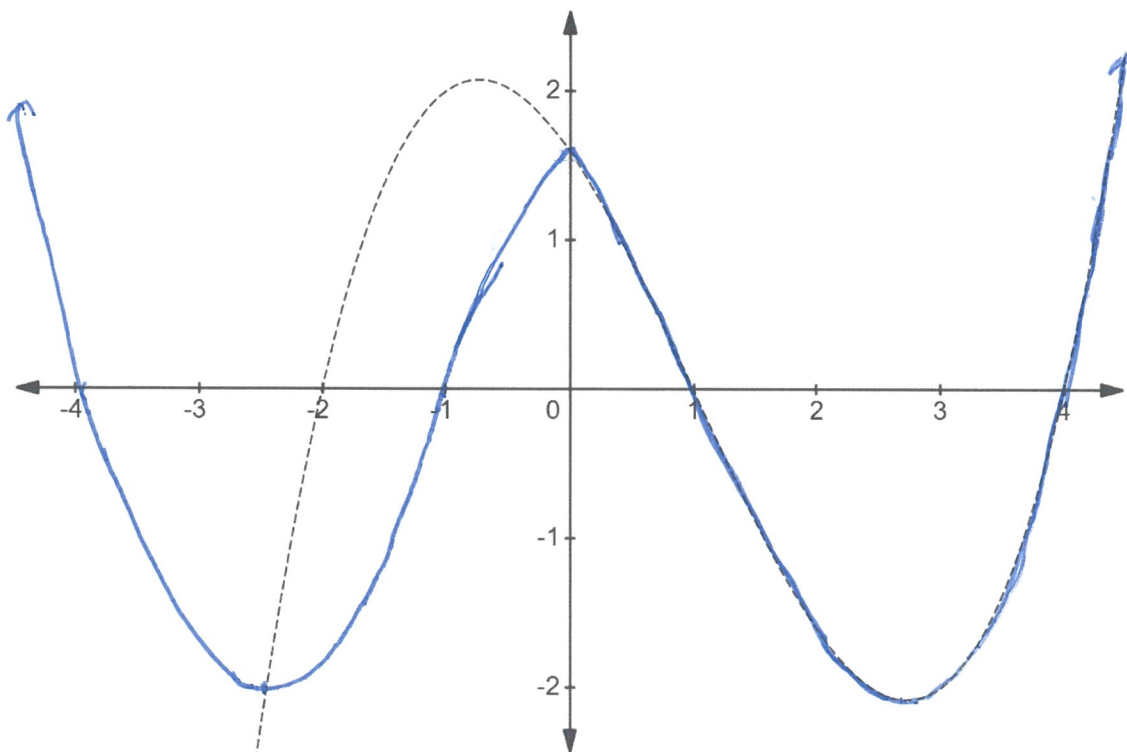
$$= \frac{32}{3} \text{ or } 10 \frac{2}{3}$$

## Question 12

(a) (i)



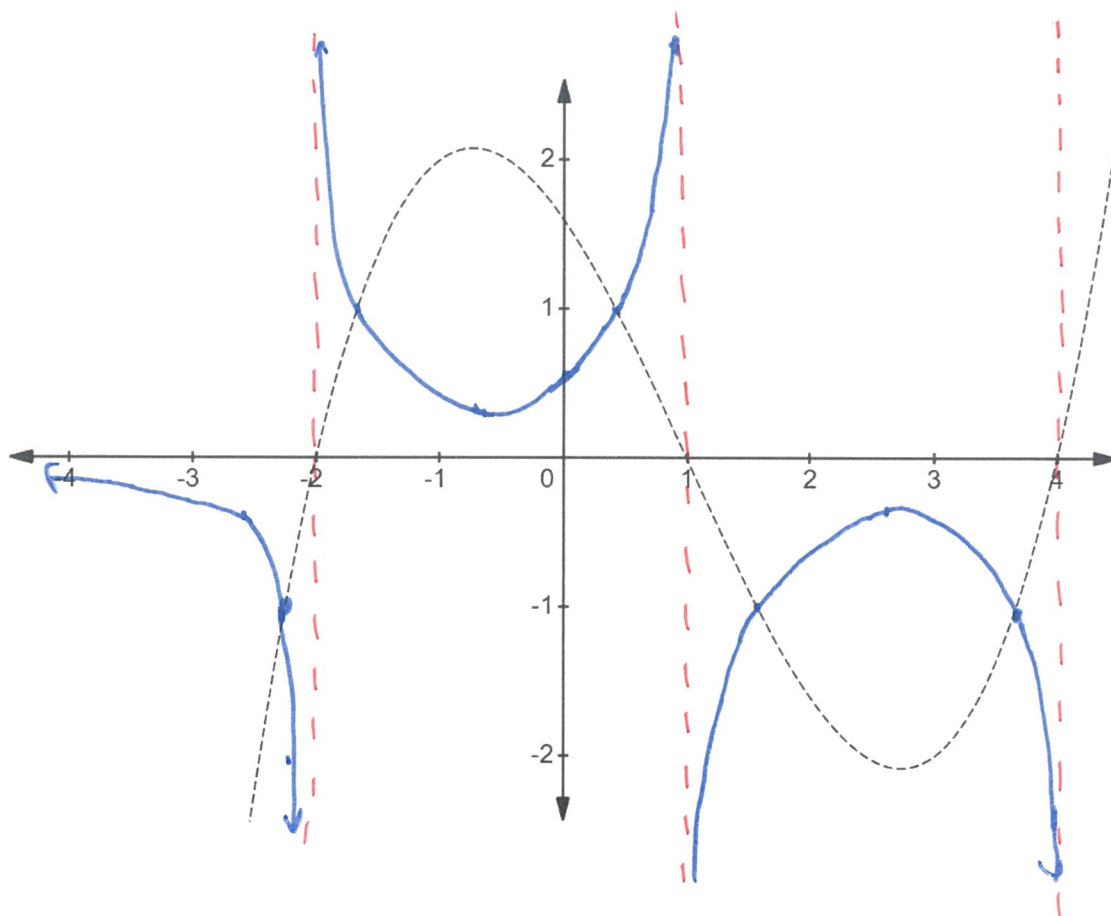
(ii)



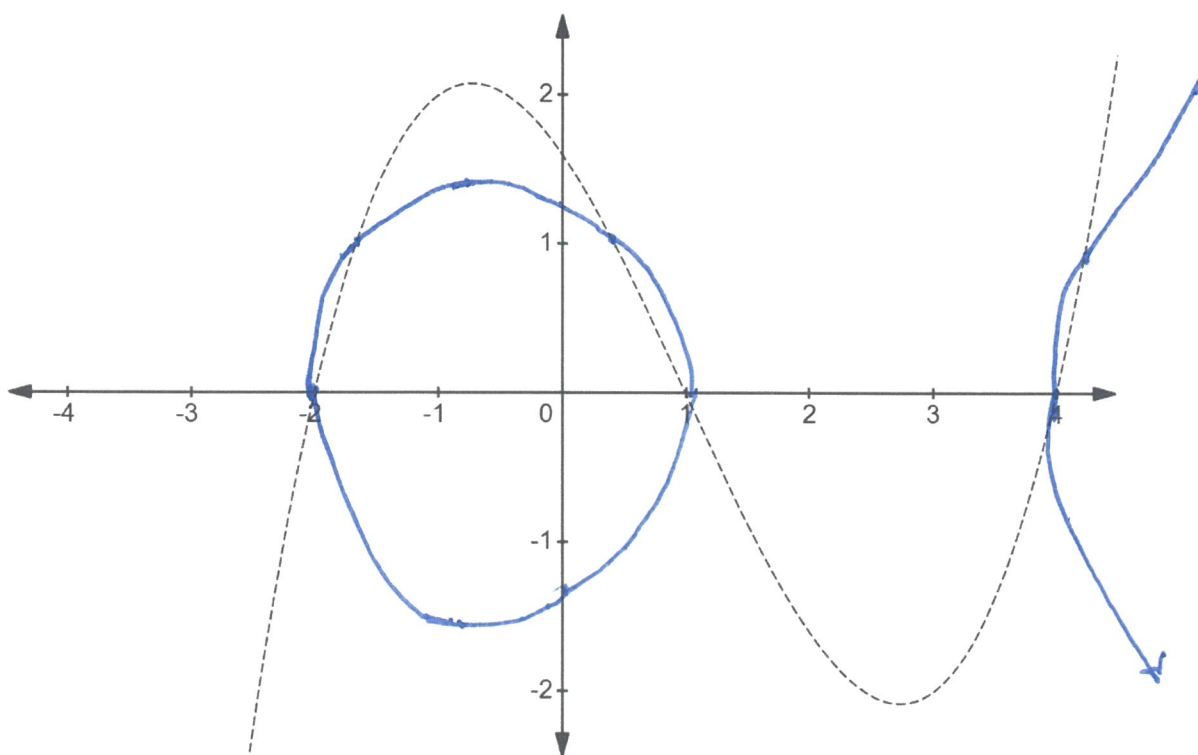
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DO NOT WRITE OUTSIDE BORDERS

(iii)

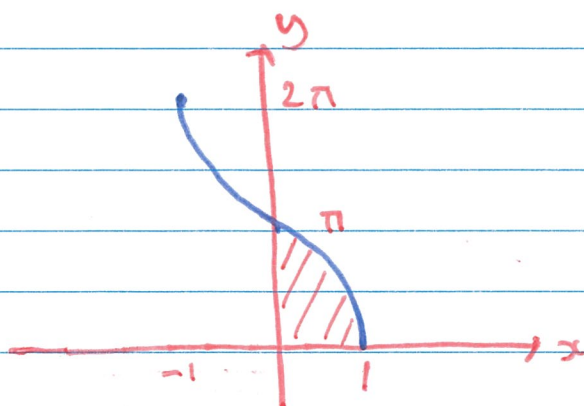


(iv)



$$(b) \underset{\substack{\uparrow \\ \text{Harry}}}{12} \times \underset{\substack{\uparrow \\ \text{Kim}}}{5} \times \underset{\substack{\uparrow \\ \text{Errand} \\ \text{Boyle} \\ \text{Titus}}}{6} p_3 \times \underset{\substack{\uparrow \\ \text{the} \\ \text{rest}}}{7!} = 36288000$$

(c) (i)



$$(ii) y = 2 \cos^{-1} x$$

$$\cos^{-1}(x) = \frac{y}{2}$$

$$x = \cos\left(\frac{y}{2}\right)$$

$$\therefore \int_0^{\pi} \cos\left(\frac{y}{2}\right) dy = \left[ 2 \sin \frac{y}{2} \right]_0^{\pi}$$

$$= 2 \sin\left(\frac{\pi}{2}\right) - \left(2 \sin\left(\frac{0}{2}\right)\right)$$

$$= 2(1)$$

$$= 2 \text{ units}^2$$

$$(iii) \pi \int_0^{\pi} \cos^2\left(\frac{y}{2}\right) dy$$

$$= \pi \int_0^{\pi} \left( \frac{1}{2} + \frac{1}{2} \cos y \right) dy$$

$$= \pi \left[ \frac{y}{2} + \frac{1}{2} \sin y \right]_0^{\pi}$$

$$= \pi \left[ \left( \frac{\pi}{2} + \frac{1}{2} \sin \pi \right) - \left( \frac{0}{2} + \frac{1}{2} \sin 0 \right) \right]$$

$$= \frac{\pi^2}{2} \text{ units}^3$$

## Question 13

(a) Base Case ( $n = 1$ )

$$\begin{aligned}1^3 + (1+1)^3 + (2+1)^3 &= 1 + 8 + 27 \\&= 36 \\&= 9(4)\end{aligned}$$

$\therefore$  true for  $n = 1$

Assume true for  $n = k$

$$\begin{aligned}\text{Assume } k^3 + (k+1)^3 + (k+2)^3 &= 9P, \quad k, P \in \mathbb{Z} \\ \therefore (k+1)^3 + (k+2)^3 &= 9P - k^3 \quad * * *\end{aligned}$$

Prove true for  $n = k+1$

$\text{RTP: } (k+1)^3 + (k+2)^3 + (k+3)^3 = 9M, \quad m \in \mathbb{Z}$

$$\begin{aligned}\text{LHS} &= (k+1)^3 + (k+2)^3 + (k+3)^3 \\&= 9P - k^3 + (k+3)^3 \quad (\text{from } * * *) \\&= 9P - k^3 + k^3 + \binom{3}{1}k^2(3) + \binom{3}{2}k(3)^2 + 3^3 \\&= 9P + 9k^2 + 27k + 27 \\&= 9(P + k^2 + 3k + 3) \\&= 9M \quad (M = P + k^2 + 3k + 3)\end{aligned}$$

$\therefore$  true for  $n = k+1$

$\therefore$  as the statement is true for  $n = 1$  and will be true for  $n = k+1$  if it is true for  $n = k$ , it is true for all positive integers.

$$(b)(i) \quad (1)^3 - 4(1)^2 + 5(1) - 2 = 0 \quad \therefore (x-1) \text{ is a factor}$$

$$(2)^3 - 4(2)^2 + 5(2) - 2 = 0 \quad \therefore (x-2) \text{ is a factor}$$

$$\begin{array}{r} x-1 \\ x^2-3x+2 \overline{) x^3-4x^2+5x-2} \\ \underline{x^3-3x^2+2x} \phantom{-2} \downarrow \\ -x^2+3x-2 \\ \underline{-x^2+3x-2} \\ 0 \end{array}$$

$$\therefore x^3 - 4x^2 + 5x - 2 = (x-1)^2(x-2)$$

$$\therefore x = 1 \text{ or } x = 2$$

$$(ii) \quad 2 \sin^2 \theta - \sin \theta \sec \theta = 2 \sin 2\theta - 2$$

$$2 \sin^2 \theta - \frac{\sin \theta}{\cos \theta} = 4 \sin \theta \cos \theta - 2 \quad \left( \div \cos^2 \theta \right)$$

$$2 \frac{\sin^2 \theta}{\cos^2 \theta} - \frac{\sin \theta}{\cos^2 \theta} = \frac{4 \sin \theta \cos \theta}{\cos^2 \theta} - \frac{2}{\cos^2 \theta}$$

$$\therefore 2 \tan^2 \theta - \tan \theta \sec^2 \theta = 4 \tan \theta - 2 \sec^2 \theta$$

$$2 \tan^2 \theta - \tan \theta (1 + \tan^2 \theta) = 4 \tan \theta - 2(1 + \tan^2 \theta)$$

$$(iii) \quad 2 \tan^2 \theta - \tan \theta - \tan^3 \theta = 4 \tan \theta - 2 - 2 \tan^2 \theta$$

$$\therefore \tan^3 \theta - 4 \tan^2 \theta + 5 \tan \theta - 2 = 0$$

$$\therefore (\tan \theta - 1)^2 (\tan \theta - 2) = 0$$

$$\therefore \tan \theta = 1 \quad \text{or} \quad \tan \theta = 2$$

$$\theta = \pi/4 \quad \theta = \tan^{-1}(2)$$

$$\begin{aligned}
 \text{(c) (i) } \vec{OX} &= \vec{OA} + \frac{1}{2} \vec{AC} \\
 &= \underline{a} + \frac{1}{2} (\underline{c} - \underline{a}) \\
 &= \underline{a} + \frac{1}{2} \underline{c} - \frac{1}{2} \underline{a} \\
 &= \frac{1}{2} \underline{a} + \frac{1}{2} \underline{c}
 \end{aligned}$$

$$\text{(ii) } \frac{|\vec{OB}|}{|\vec{OC}|} = \frac{k+1}{k}$$

$$|\vec{OB}| = |\vec{OC}| \left( \frac{k+1}{k} \right)$$

$$\therefore \vec{OB} = \underline{c} \left( 1 + \frac{1}{k} \right)$$

$$\text{(iii) } \vec{XD} = \vec{OB} - \vec{OX}$$

$$3\underline{c} - \frac{1}{2}\underline{a} = \underline{c} \left( 1 + \frac{1}{k} \right) - \frac{1}{2}(\underline{a} + \underline{c})$$

$$3\underline{c} - \cancel{\frac{1}{2}\underline{a}} = \underline{c} + \frac{1}{k}\underline{c} - \cancel{\frac{1}{2}\underline{a}} - \frac{1}{2}\underline{c}$$

$$3\underline{c} = \frac{1}{2}\underline{c} + \frac{1}{k}\underline{c}$$

$$\frac{5}{2}\underline{c} = \frac{1}{k}\underline{c}$$

$$\frac{2}{5}\underline{c} = k\underline{c}$$

$$\therefore k = \frac{2}{5}$$

$$\begin{aligned}
 (d)(i) \text{ LHS} &= \binom{n+1}{k+1} \\
 &= \frac{(n+1)!}{(k+1)! [(n+1)-(k+1)]!} \\
 &= \frac{(n+1)n!}{(k+1)k!(n-1-k+1)!} \\
 &= \frac{n+1}{k+1} \times \frac{n!}{k!(n-k)!} \\
 &= \frac{n+1}{k+1} \binom{n}{k} \\
 &= \text{RHS}
 \end{aligned}$$

$$\begin{aligned}
 (ii) (1+x)^{n+1} &= \binom{n+1}{0} + \binom{n+1}{1}x + \binom{n+1}{2}x^2 + \dots + \binom{n+1}{n+1}x^{n+1} \\
 &= 1 + \binom{n+1}{1}x + \binom{n+1}{2}x^2 + \dots + \binom{n+1}{n+1}x^{n+1}
 \end{aligned}$$

$$\begin{aligned}
 \therefore (1+x)^{n+1} - 1 &= \binom{n+1}{1}x + \binom{n+1}{2}x^2 + \dots + \binom{n+1}{n+1}x^{n+1} \\
 &= \frac{n+1}{1} \binom{n}{0}x + \frac{n+1}{2} \binom{n}{1}x^2 + \dots + \frac{n+1}{n+1} \binom{n}{n}x^{n+1} \quad (\text{from (i)}) \\
 &= x(n+1) \left[ \frac{1}{1} \binom{n}{0} + \frac{1}{2} \binom{n}{1}x + \dots + \frac{1}{n+1} \binom{n}{n}x^n \right]
 \end{aligned}$$

$$\therefore \frac{(1+x)^{n+1} - 1}{x(n+1)} = \binom{n}{0} + \frac{1}{2} \binom{n}{1}x + \dots + \frac{1}{n+1} \binom{n}{n}x^n$$

$$\therefore \binom{n}{0} + \frac{1}{2} \binom{n}{1}x + \frac{1}{3} \binom{n}{2}x^2 + \frac{1}{4} \binom{n}{3}x^3 + \dots + \frac{1}{n+1} \binom{n}{n}x^n = \frac{(1+x)^{n+1} - 1}{x(n+1)}$$

## Question 14

(a) let 2 numbers have the form  $2^{a_1} 3^{a_2} 5^{a_3} 7^{a_4}$  and  $2^{b_1} 3^{b_2} 5^{b_3} 7^{b_4}$

$$\text{product} = 2^{a_1+b_1} 3^{a_2+b_2} 5^{a_3+b_3} 7^{a_4+b_4}$$

$a_1 + b_1, a_2 + b_2, a_3 + b_3$  &  $a_4 + b_4$  must all be even

$\therefore a_1 \& b_1, a_2 \& b_2, a_3 \& b_3, a_4 \& b_4$  must have the same parity

there are  $2 \times 2 \times 2 \times 2 = 16$  possible parity permutations for each number

$\therefore$  by pigeonhole principle  $16 + 1 = 17$  numbers are needed.

(b)(i)  $y(0) \cdot y(t) = 0$

$$(U \cos \theta \mathbf{i} + (U \sin \theta - gt) \mathbf{j}) \cdot (U \cos \theta \mathbf{i} + (U \sin \theta - gt) \mathbf{j}) = 0$$

$$U^2 \cos^2 \theta + U \sin \theta (U \sin \theta - gt) = 0$$

$$U^2 \cos^2 \theta + U^2 \sin^2 \theta - U \sin \theta gt = 0$$

$$U^2 (\sin^2 \theta + \cos^2 \theta) = U \sin \theta gt$$

$$U^2 = U \sin \theta gt$$

$$\therefore t = \frac{U}{g \sin \theta}$$

(ii) at P ( $r(t) = Ut \cos \theta$ )

$$p = Ut \cos \theta$$

$$p = U \left( \frac{U}{g \sin \theta} \right) \cos \theta$$

$$p = \frac{U^2 \cos \theta}{g \sin \theta}$$

$$U^2 = \frac{p g \sin \theta}{\cos \theta}$$

$$\therefore U^2 = p g \tan \theta$$

$$U = \sqrt{p g \tan \theta} \quad (U > 0)$$

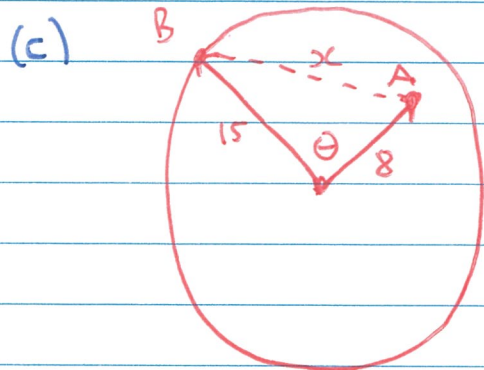
$$\begin{aligned}
 \text{(iii)} \quad W^2 &= (U \cos \theta)^2 + (U \sin \theta - gt)^2 \\
 &= (U \cos \theta)^2 + \left( U \sin \theta - \cancel{g} \times \frac{U}{\cancel{g} \sin \theta} \right)^2 \quad (\text{from (i)}) \\
 &= U^2 \cos^2 \theta + U^2 \sin^2 \theta - 2U^2 + \frac{U^2}{\sin^2 \theta} \\
 &= U^2 \left( \cos^2 \theta + \sin^2 \theta - 2 + \frac{1}{\sin^2 \theta} \right) \\
 &= U^2 (1 - 2 + \operatorname{cosec}^2 \theta) \\
 &= U^2 (\operatorname{cosec}^2 \theta - 1) \\
 &= U^2 \cot^2 \theta
 \end{aligned}$$

$$\therefore W = U \cot \theta \quad (W, U, \cot \theta > 0 \text{ for } 0 < \theta < 90^\circ)$$

$$\therefore UW = U^2 \cot \theta$$

$$= pg \tan \theta \cancel{\cot \theta}$$

$$= pg \quad \text{which is independent of } g \text{ \& } \theta$$



$$x = \sqrt{15^2 + 8^2 - 2 \times 15 \times 8 \times \cos \theta}$$

$$= (289 - 240 \cos \theta)^{\frac{1}{2}}$$

$$\frac{dx}{d\theta} = \left(\frac{1}{2}\right) (240 \sin \theta) (289 - 240 \cos \theta)^{-\frac{1}{2}}$$

$$= \frac{120 \sin \theta}{\sqrt{289 - 240 \cos \theta}}$$

$$\frac{d\theta}{dt} = \frac{2\pi}{60} - \frac{2\pi}{12 \times 60}$$

$$= \frac{11\pi}{360}$$

$$\frac{dx}{dt} = \frac{dx}{d\theta} \times \frac{d\theta}{dt}$$

$$= \frac{120 \sin \theta}{\sqrt{289 - 240 \cos \theta}} \times \frac{11\pi}{360}$$

$$\boxed{\text{when } \theta = \frac{\pi}{2}}$$

$$= \frac{\cancel{120} \sin(\frac{\pi}{2})}{\sqrt{\cancel{289} - \cancel{240} \cos \theta}} \times \frac{11\pi}{\cancel{360} 3}$$

$$= \frac{11\pi}{51} \text{ cm/min}$$

(d) Note that one root must be a double root.

Let the roots be  $r, r, s$  &  $t$

$$r + r + s + t = r^2s + r^2t + rst + rst \quad (\text{both } = \frac{a}{t})$$

$$2r + s + t = r^2s + r^2t + 2rst$$

$$2r - 2rst = r^2s + r^2t - s - t$$

$$2r(1-st) = r^2(s+t) - 1(s+t)$$

$$2r(1-st) = (s+t)(r^2-1)$$

$\circledast$

$$\therefore \frac{2r}{r^2-1} = \frac{s+t}{1-st}$$

$$\therefore -\frac{2r}{1-r^2} = \frac{s+t}{1-st}$$

$$\boxed{\text{let } r = \tan 3\theta}$$

$$-\frac{2\tan 3\theta}{1-\tan^2 3\theta} = \frac{\tan \theta + \tan 2\theta}{1 - \tan \theta \tan 2\theta}$$

$$-\tan 6\theta = \tan 3\theta$$

$$\therefore \tan(-6\theta) = \tan 3\theta$$

$$\therefore 3\theta - (-6\theta) = 0, \pi, 2\pi \dots \quad (\text{period} = \pi)$$

$$\therefore 9\theta = \pi \quad (\text{smallest value for distinct roots})$$

$$\therefore \theta = \frac{\pi}{9}$$

note 1 if we take  $r = \tan \theta$  we get  $\theta = \frac{\pi}{7}$

$$r = \tan 2\theta \Rightarrow \theta = \frac{\pi}{8}$$

so all 3 cases should be checked

note 2 to ensure we are not dividing by 0

in  $\circledast$  we need to check when

$$r^2 - 1 = 0$$

$$\therefore r^2 = 1$$

case 1  $\tan^2 \theta = 1$

$$\theta = \pi/4$$

$$\therefore \tan 2\theta = \tan(\pi/2) \text{ which is undefined}$$

$\therefore$  not a solution

case 2  $\tan^2 2\theta = 1$

$$2\theta = \pi/4$$

$$\theta = \pi/8 \text{ (already found)}$$

case 3  $\tan^2 3\theta = 1$

$$3\theta = \pi/4$$

$$\theta = \pi/12$$

Looks good - smaller than  $\pi/9$  but

$$\text{then } \tan 2\theta = \pi/6 \text{ \& } \tan 3\theta = \pi/4$$

subbing into LHS of  $\textcircled{*}$

$$2 \tan(\pi/12) (1 - \tan(\pi/6) \tan(\pi/4)) = 0.2264973 \dots \neq 0$$

$\therefore \theta = \pi/12$  is not a solution

$$\theta = \pi/9 \text{ is the answer}$$